

- 3 (a) increasing separation of molecules / breaking bonds between molecules B1  
 (allow atoms/molecules, overcome forces)  
 doing work against atmosphere (during expansion) B1 [2]
- (b) (i) 1 either bubbles produced at a constant rate / mass evaporates/lost at  
constant rate  
 or find mass loss more than once and this rate should be constant B1 [1]  
 or temperature of liquid remains constant B1 [1]
- 2 to allow/cancel out/eliminate/compensate for heat losses (to atmosphere) B1 [1]  
 (do not allow 'prevent'/'stop')
- (ii) use of  $power \times time = mass \times specific\ latent\ heat$  C1  
 $(70 - 50) \times 5 \times 60 = (13.6 - 6.5) \times L$  C1  
 $L = 845 \text{ J g}^{-1}$  A1 [3]

|               |                                       |                 |              |
|---------------|---------------------------------------|-----------------|--------------|
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|               | <b>GCE A/AS LEVEL – May/June 2009</b> | <b>9702</b>     | <b>04</b>    |

4. (c) (i)  $(0 =) \dots t$  (allow any subject if all terms given)

D1 [1]